United International University
Department of Computer Science and Engineering

CSE 2213/CSI 219: Discrete Mathematics (BSCSE and BSDS)

Final-term Examination Summer 2025

Total Marks: 40 Time: 2 hour

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. Prove the following using Mathematical Induction, whenever n is a positive integer [3]

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n! = (n+1)! - 1$$

2. (a) A group of friends is organizing a "Netflix Night" where everyone picks movies to watch. There are 10 genres available on Netflix (Action, Comedy, Romance, Thriller, Sci-Fi, Mystery, Horror, Fantasy, Documentary, and Animation). Each person picks one genre as their favorite for the night. How many people must attend the event to guarantee that at least 4 people pick the same genre, no matter how the choices are made? [3]

   (b) A university's Computer Science department has 8 female and 10 male faculty members. In how many ways can a 7 member committee be formed if at least three women and at least two men must be included in the committee? [3]

   (c) A website offers users two options for creating a password: [2*2=4]

   i. The password has 3 uppercase letters (A–Z) followed by 2 digits (0–9) and 1 special symbol (chosen from @, #, $, %, &).

   ii. The password has 4 uppercase letters (A–Z) and 2 special symbols (chosen from @, #, $, %, &).

   Each password must contain at least one special symbol. Find the total number of possible passwords that can be created if a user can choose either option.

3. (a) Dhaka Metro Rail has several stations connected by train lines. Some pairs of stations are connected by several lines, where each line operates in both directions. No train starts and ends at the same station. Each station is treated as a vertex, and each train line as an edge.

   i) What kind of graph best represents the Dhaka Metro Rail network? [0.5]

   ii) For testing or maintenance, a train track is returned to the same station, allowing trains to run in circles for diagnostics. What type of graph best represents this temporary maintenance network? [0.5]

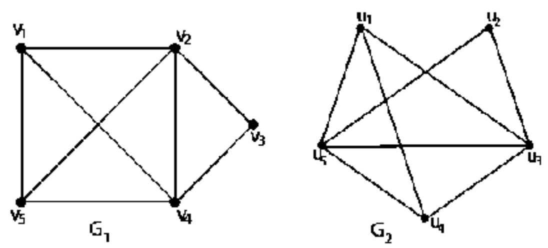
   (b) If C_n has n vertices and 18 edges, find the value of n. [2]

   (c) An undirected graph has 10 vertices and 17 edges. One vertex has degree d. The degrees of the other 9 vertices are: $4, 3, 2, 2, 2, 1, 1, 1, 1$.

   i) Using the Handshaking Theorem, find the value of d. [1]

   ii) What type of graph could this be? Explain your reasoning. [1]

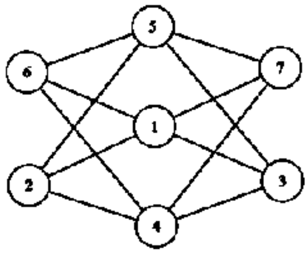
   (d) Determine whether the given pair of graphs is isomorphic. If they are, give a function f that defines the isomorphism. If they are not, give an invariant for graph isomorphism that they do not share. [3]



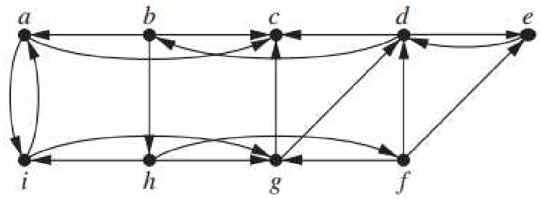
4. (a) Draw the undirected graph from the following incidence matrix. [3]

	$e1$	$e2$	$e3$	$e4$	$e5$	$e6$	$e7$	$e8$
a	1	1	0	0	0	0	1	0
b	0	0	0	1	1	0	0	0
c	0	0	1	0	1	1	0	1
d	0	0	1	0	0	0	0	0
e	0	1	0	1	0	1	1	0

(b) Determine whether the graph in the following figure is bipartite or not, using the 2-coloring algorithm. If the graph is bipartite, redraw the graph in bipartite form. [3]



(c) Determine whether the following graph is strongly connected or weakly connected. If it is weakly connected, identify the strongly connected components. [3]



5. (a) Consider the following words: **error, count, truck, about, zidane, dollar, beach, zico**.

   i. Construct a Binary Search Tree (BST) by inserting them one by one in the given order. [3]

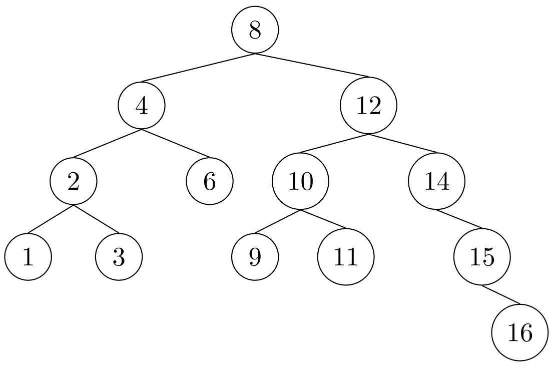
   ii. Write the preorder, inorder, and postorder traversal sequences for the binary search tree. [3]

   (b) Answer the following:

   (i) For a full 4-ary tree with 10 internal vertices, how many leaves does it have? [1]

   (ii) For a 5-ary tree of height 4, what is the maximum number of leaves? [1]

   (c) Consider the following binary tree:



   (i) State the height of the above tree. **[1]**

   (ii) For the node 2, list all of its ancestors and all of its descendants. **[1]**